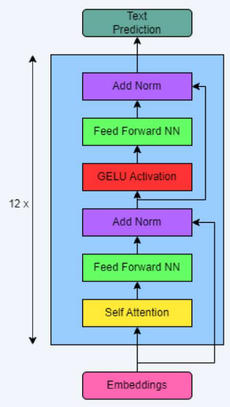
MODELING PERFORMANCE OF TRANSFORMERS ON FPGA

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ABSTRACT:

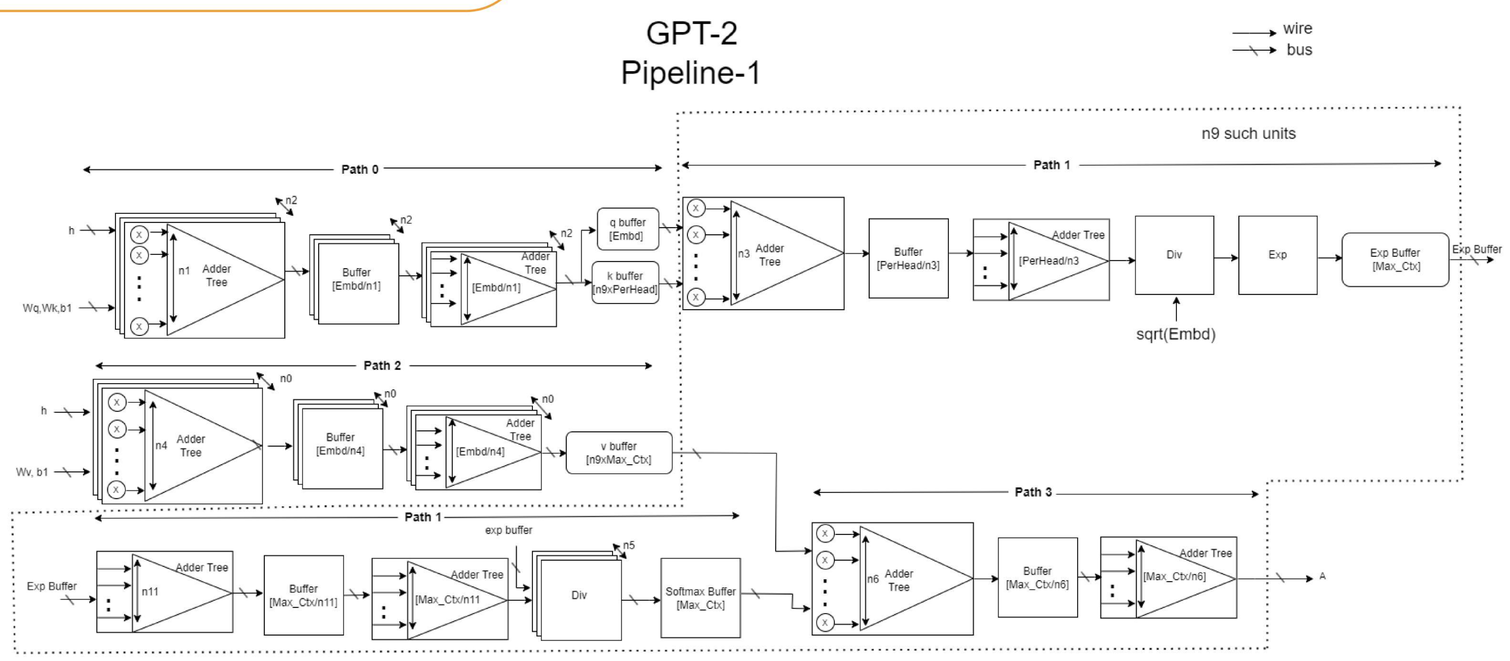
The surge of transformer models like GPT and BERT has transformed AI applications. Deploying these on edge devices poses challenges. Our paper, using GPT-2 as a reference, proposes an **optimal FPGA deployment** strategy. We explore two designs, demonstrating **21x** and **119x** better latency than a dual-core i5 CPU, considering the model-agnostic and model-specific approaches.



DESIGN APPROACH:

- Parameterized model-specific architecture for **GPT-2** — A decoder-only LLM by OpenAI
- Used off-the-shelf **Adder**, **Multiplier**, exponential and other special function circuits
- Optimal partitioning of resources to minimize latency using **Genetic Algorithm**

SELF ATTENTION BLOCK



LEGEND

T_{mul} = Multiplier Delay
T_{add} = Adder Delay
T_{buffer} = Buffer
h = Input Embeddings
W, b = Weights & Biases
A = Self Attention

LATENCY FORMULATION:

$$IL_1 = T_{mul} + \lceil \log(n_1) \rceil * T_{add}$$

$$IL_2 = \log \left\lceil \frac{768}{n_1} \right\rceil * T_{add}$$

$$IL_{2(1)} = T_{mul} + \lceil \log(n_3) \rceil * T_{add}$$

$$IL_3 = \log \left\lceil \frac{64}{n_3} \right\rceil * T_{add} + T_{div} + T_{exp}$$

$$IL_4 = \lceil \log(n_{13}) \rceil * T_{add}$$

$$IL_5 = \log \left\lceil \frac{1024}{n_{13}} \right\rceil * T_{add} + T_{div}$$

$$IL_{5(1)} = T_{mul} + \lceil \log(n_6) \rceil * T_{add}$$

$$IL_6 = \log \left\lceil \frac{1024}{n_6} \right\rceil * T_{add} + T_{mul} + \lceil \log(n_{10}) \rceil * T_{add}$$

$$IL_7 = \log \left\lceil \frac{768}{n_{10}} \right\rceil$$

$$IL_{path0} = IL_1 + \left\lceil \frac{768}{n_1} \right\rceil * T_{buffer} + IL_2$$

$$IL_{path1} = IL_{2(1)} + IL_3 + IL_4 + IL_5 + \left(\left\lceil \frac{64}{n_3} \right\rceil + 1024 + \left\lceil \frac{1024}{n_{13}} \right\rceil \right) * T_{buffer}$$

$$IL_{path2} = T_{mul} + \left(\lceil \log(n_4) \rceil + \log \left\lceil \frac{768}{n_4} \right\rceil \right) * T_{add} + \left\lceil \frac{768}{n_4} \right\rceil * T_{buffer} - (IL_1 + IL_2 + IL_3 + IL_4)$$

$$IL_{path3} = \left(\left\lceil \frac{64}{n_6} \right\rceil + \left\lceil \frac{768}{n_{10}} \right\rceil + 768 \right) * T_{buffer} + IL_{5(1)} + IL_6 + IL_7$$

$$Latency = max \left(Il_{path0} + \frac{768 * 768 * 1024}{n_1 * n_2} + IL_{path1} + \frac{64 * 1024}{n_3}, IL_{path2} + \frac{768 * 1024 * 64}{n_4 * n_{10}} \right) + IL_{path3} + 1024 * \frac{12 * 1024}{n_{10}}$$

RESULTS:

- Latencies of both Design1(Model Agnostic) and Design2(Model Specific) computed for Xilinx Alveo-U280
- Design1 gave a latency of **1.75s** and Design2 gave **0.316s** and a dual-core CPU gave a latency of **37.69s**

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